

GUIDED LECTURE NOTES
for
PROOF, SET THEORY, AND LOGIC
MATH 330

Part III: Real Numbers and Cardinality

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These guided notes accompany the Fall 2026 revised course edition of *Elementary Set Theory: Proof, Set Theory, and Logic*. They are based on Richard P. Millspaugh's original textbook materials and lecture materials prepared and revised by Bryce A. Christopherson.

The blank spaces in the student copy are completed during class. Each blank occupies exactly the same space as the corresponding typed completion in the instructor copy, so the two packets have identical pagination. Definitions, theorem statements, and the surrounding explanations are provided so that class time can focus on examples, proof decisions, and the mathematical steps that connect them.

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CHAPTER 6

THE REAL NUMBERS

INTRODUCTION

We now turn our attention to developing a mathematical description of the real numbers. All of the results of calculus can be derived from these basic properties of the real number system, which is usually done in a first course in real analysis. Our basic assumptions about the real numbers are called axioms, and they are divided into several groups. Many of these axioms will be familiar to you from previous courses in algebra.

6.1 FIELD AXIOMS

Taken as a group, the field axioms tell us that the real numbers together with the operations of addition and multiplication form what is known as a *field*. Very informally, you might think of a field as something that satisfies the usual rules you encountered in high school algebra. Fields and related objects are studied in more depth in a course in abstract algebra.

Field Axioms.

The set of real numbers $\mathbb{R}$ is a collection of objects together with the two binary operations addition and multiplication that satisfy the following properties (known as the field axioms):

- (i) Commutative Laws: For every $a, b \in \mathbb{R}$, $a + b = b + a$ and $ab = ba$.
- (ii) Associative Laws: For every $a, b, c \in \mathbb{R}$, $(a + b) + c = a + (b + c)$ and $(ab)c = a(bc)$.
- (iii) Distributive Law: For every $a, b, c \in \mathbb{R}$, $a(b + c) = ab + ac$.
- (iv) Identities: There are distinct elements 0 and 1 in $\mathbb{R}$ such that $a \cdot 1 = a$ and $a + 0 = a$ for every $a \in \mathbb{R}$.
- (v) Additive Inverses: For every $a \in \mathbb{R}$, there is an element $-a \in \mathbb{R}$ such that $a + (-a) = 0$.
- (vi) Multiplicative Inverses: For every $a \in \mathbb{R}$ with $a \neq 0$, there is an element $a^{-1} \in \mathbb{R}$ such that $aa^{-1} = 1$.

Any set of objects satisfying all of these properties is a *field*. The rational numbers $\mathbb{Q}$ and complex numbers $\mathbb{C}$ are also fields. The additional order and completeness axioms below distinguish $\mathbb{R}$ from these examples.

A number of the familiar algebraic properties of the real numbers are immediate consequences of the field axioms. A few of them are collected in the following theorem, though there are certainly many others.

THEOREM 6.1.

For any real numbers a, b, c , the following are true:

- (i) *If $a + b = a + c$, then $b = c$.*
- (ii) *$-(-a) = a$.*
- (iii) *If $a \neq 0$ and $ab = ac$, then $b = c$.*
- (iv) *If $a \neq 0$, then $(a^{-1})^{-1} = a$.*
- (v) *$a \cdot 0 = 0$.*
- (vi) *$-a = (-1)a$.*
- (vii) *If $ab = 0$, then $a = 0$ or $b = 0$.*

Proof.

We prove parts (ii), (iii), and (vi). The remaining proofs are exercises.

We first show that (ii) is true. To see this, note that $-(-a)$ is the additive inverse of $-a$. The reader should justify each of the following steps:

$$\begin{aligned} a + (-a) &= 0 \\ &= -a + (-(-a)) \\ &= -(-a) + (-a) \end{aligned}$$

so $a + (-a) = -(-a) + (-a)$. We apply part (i) to see that $a = -(-a)$ as desired.

To prove (iii), suppose that $a \neq 0$ and $ab = ac$. Since $a \neq 0$, a has a multiplicative inverse a^{-1} . The reader should give a justification for each step of the following:

$$\begin{aligned} b &= 1 \cdot b \\ &= (a^{-1}a)b \\ &= a^{-1}(ab) \\ &= a^{-1}(ac) \\ &= (a^{-1}a)c \\ &= 1 \cdot c \\ &= c \end{aligned}$$

To prove (vi), let a be any real number. Then:

$$\begin{aligned} a + (-1)a &= 1 \cdot a + (-1)a \\ &= a(1 + (-1)) \\ &= a \cdot 0 \\ &= 0 \end{aligned}$$

Hence $a + (-1)a = 0 = a + (-a)$ and we may apply part (i) to see that $(-1)a = -a$.

6.2 ORDER AXIOMS

As noted previously, there are a number of common fields. One of the differences between $\mathbb{R}$ and $\mathbb{C}$ is that we think of the real numbers as lying in order along a line. There is no natural way to organize the complex numbers in this fashion. The next group of axioms make precise what we mean when we say that the real numbers form an *ordered field*.

Order Axioms. There is an order $<$ defined on the real numbers $\mathbb{R}$ satisfying:

(i) Transitivity: If $a < b$ and $b < c$, then $a < c$.

(ii) Trichotomy: For every two real numbers a and b , exactly one of the following holds:

$$a < b \quad \text{or} \quad a = b \quad \text{or} \quad b < a$$

(iii) If $a < b$, then $a + c < b + c$ for every $c \in \mathbb{R}$.

(iv) If $a < b$ and $c > 0$, then $ac < bc$.

We derive several consequences of the fact that $\mathbb{R}$ is an ordered field.

THEOREM 6.2.

The following statements are true in $\mathbb{R}$:

- (i) $a > 0$ iff $-a < 0$.
- (ii) If $a < b$, then $-a > -b$.
- (iii) If $a \neq 0$, then $a^2 > 0$.
- (iv) $1 > 0$.

Proof.

To prove part (i), assume first that $a > 0$. We have:

$$\begin{aligned}
 a &> 0 \\
 a + (-a) &> 0 + (-a) && \text{using Order Axiom 3} \\
 0 &> -a
 \end{aligned}$$

The converse is similar and left as an exercise.

We next prove part (ii). The reader should determine which axiom or theorem justifies each step of the following:

$$\begin{aligned}
 a &< b \\
 a + (-b) &< b + (-b) \\
 a + (-b) &< 0 \\
 ((-a) + a) + (-b) &< (-a) + 0 \\
 -b &< -a
 \end{aligned}$$

The proofs of the remaining parts of the theorem are left to the reader.

THEOREM 6.3.

Let $a, b \in \mathbb{R}$.

- (i) If $a > 0$ and $b > 0$, then $ab > 0$.
- (ii) If $a < 0$ and $b < 0$, then $ab > 0$.
- (iii) If $a > 0$ and $b < 0$, then $ab < 0$.

THEOREM 6.4.

Let $a \in \mathbb{R}$.

- (i) If $a > 0$, then $a^{-1} > 0$.
- (ii) If $a < 0$, then $a^{-1} < 0$.

Proof.

To prove part (i) we assume $a > 0$ and apply Trichotomy. If $a^{-1} < 0$, then $1 = aa^{-1} < 0$ by Theorem 6.3, but this contradicts Theorem 6.2, part (iv). If $a^{-1} = 0$, then we have $1 = aa^{-1} = a \cdot 0 = 0$, which again contradicts Theorem 6.2, part (iv). The only remaining possibility is that $a^{-1} > 0$ as desired.

The proof of (ii) is similar and is left for the reader.

THEOREM 6.5.

Let $a \geq 0$ and $b \geq 0$. Then $a < b$ iff $a^2 < b^2$.

Proof.

Assume $a, b \geq 0$. If $a < b$, then $b - a > 0$ and $a + b > 0$, so

$$b^2 - a^2 = (b - a)(a + b) > 0.$$

Hence $a^2 < b^2$.

Conversely, suppose that $a^2 < b^2$. The possibility $a = b$ is immediately excluded. If $b < a$, the direction just proved, applied with a and b interchanged, gives $b^2 < a^2$, also a contradiction. Trichotomy therefore forces $a < b$.

6.3 COMPLETENESS OF THE REAL NUMBERS

Our axioms so far ensure that $\mathbb{R}$ is an ordered field. The field $\mathbb{Q}$ of rational numbers is also an ordered field, so we need something further to distinguish between $\mathbb{R}$ and $\mathbb{Q}$. Our last axiom is based on the observation that the field $\mathbb{Q}$ has *holes*, whereas $\mathbb{R}$ does not. Let's first consider an example to clarify what we mean by this.

EXAMPLE 6.6.

Let A and B be the following sets:

$$A = \{q \in \mathbb{Q} \mid q > 0 \text{ and } q^2 \leq 2\}$$

$$B = \{x \in \mathbb{R} \mid x > 0 \text{ and } x^2 \leq 2\}$$

As we will show in Theorem 6.13, there is a real number $\sqrt{2}$ whose square is 2. The number $\sqrt{2}$ is in B and $\sqrt{2}$ is larger than every other number in B .

As we showed in Theorem 2.18, there is no rational number whose square is 2. It can be shown that for every number in A , there are larger numbers that are also in A .

6.3.1 UPPER AND LOWER BOUNDS

DEFINITION 6.7. Let A be a nonempty set of real numbers. We say that a number u is an *upper bound* for A if $x \leq u$ for every $x \in A$. We say that a number m is a *lower bound* for A if $x \geq m$ for every $x \in A$. We say that A is *bounded above* if A has an upper bound, that A is *bounded below* if A has a lower bound, and that A is *bounded* if A is bounded above and below.

EXAMPLE 6.8.

Let $A = \{a \in \mathbb{R} \mid a^2 \leq 2\}$, $B = [0, 7)$, $C = \mathbb{N}$, and $D = \mathbb{Z}$.

The number 2 is an upper bound for A and -2 is a lower bound for A .

The set B has an upper bound at 7 and a lower bound at 0.

The set C has a lower bound at 1, but no upper bound; the latter claim will follow from the Archimedean Property proved below.

The set D has no upper or lower bounds, as will also follow from that property.

The previous example illustrates several things. It is possible for a nonempty set to have both upper and lower bounds, just one bound, or no bounds at all. Upper and lower bounds may or may not be elements of the set. Finally, upper and lower bounds are not unique. Transitivity implies that if M is an upper bound for a set A , then every number larger than M is also an upper bound for A . Similarly, if m is a lower bound for A then every number smaller than m is a lower bound for A .

Consider the set $A = \{a \in \mathbb{R} \mid a^2 \leq 2\}$ from the previous example again. We claimed that 2 is an upper bound, and that is certainly true, but note that $3/2$ is also an upper bound. In some sense this smaller upper bound is a “better” bound for A because it puts a tighter restriction on the size of the elements of A . This is approximately like saying that Grand Forks is a better way to describe the location of UND than North Dakota. In this sense the best possible upper bound would be the smallest one, if there is one. In this case, A has a smallest upper bound in $\mathbb{R}$, namely $\sqrt{2}$. The analogous rational set $\{q \in \mathbb{Q} \mid q^2 \leq 2\}$ has no least upper bound in $\mathbb{Q}$. This is the basic difference between $\mathbb{Q}$ and $\mathbb{R}$ that we wish to capture in an axiom. First, we need a couple more definitions.

DEFINITION 6.9. Let A be a nonempty subset of $\mathbb{R}$. We say that a number u is a *least upper bound* (LUB) for A if both of the following are true:

- (i) For every $a \in A$, $a \leq u$.
- (ii) For every upper bound b of A , $u \leq b$.

We say that a number m is a *greatest lower bound* (GLB) for A if both of the following are true:

- (i) For every $a \in A$, $a \geq m$.
- (ii) If b is a lower bound for A , then $m \geq b$.

Note that a LUB is sometimes called a *supremum* and a GLB is sometimes called an *infimum*.

It is not hard to prove that, unlike upper bounds, a set can have only one least upper bound. This fact is made explicit in the following theorem, whose proof is left as a homework exercise.

THEOREM 6.10.

Let A be a nonempty subset of $\mathbb{R}$. If u and v are least upper bounds for A , then $u = v$.

The following theorem says that the least upper bound of a set must in some sense be “close to” the set.

THEOREM 6.11.

Let A be a nonempty subset of $\mathbb{R}$ and let u be the least upper bound for A . If $x < u$, then there is an element $a \in A$ such that $x < a$.

Proof.

Suppose that u is the least upper bound for A and that $x < u$. Assume that there is no $a \in A$ such that $x < a$, then $a \leq x$ for every $a \in A$ by Trichotomy. It follows by definition that x is an upper bound for A , but this contradicts the fact that u is the least upper bound for A since $x < u$.

We are now ready to state our final axiom, which says that $\mathbb{R}$ is *complete*.

The Completeness Axiom. Let A be a nonempty subset of $\mathbb{R}$. If A has an upper bound, then A has a least upper bound.

There are a number of consequences of the Completeness Axiom, most of which are beyond the scope of this text. We will look at only a couple of them. We begin with the fairly intuitive-seeming fact that for any real number x , there is a natural number larger than x .

THEOREM 6.12.

(Archimedean Property) If $x \in \mathbb{R}$, then there is an $n \in \mathbb{N}$ such that $n > x$.

Proof.

Suppose to the contrary that some $x \in \mathbb{R}$ satisfies $n \leq x$ for every $n \in \mathbb{N}$. Then $\mathbb{N}$ is bounded above and, by completeness, has a least upper bound m . Since $m - 1 < m$, Theorem 6.11 gives an $n \in \mathbb{N}$ with $m - 1 < n$. Thus $m < n + 1$, while $n + 1 \in \mathbb{N}$, contradicting that m is an upper bound.

We previously commented, without proof, that the Completeness Axiom would allow us to distinguish $\mathbb{R}$ from $\mathbb{Q}$. We will now make this explicit by proving that there is at least one real number that is not rational.¹ By Theorem 2.18, there is no rational number whose square is 2. We now show that the Completeness Axiom implies that there must be a real number x with $x^2 = 2$.

THEOREM 6.13.

There is a number $x \in \mathbb{R}$ such that $x^2 = 2$.

Proof.

Let

$$A = \{a \in \mathbb{R} \mid a \geq 0 \text{ and } a^2 \leq 2\}.$$

¹In fact there are more irrational numbers than there are rational numbers, as we shall see in the final chapter.

The set A is nonempty because $1 \in A$. It is bounded above by 2: if $a > 2$, then $a^2 > 4 > 2$, so $a \notin A$. Completeness therefore gives a least upper bound $x = \sup A$. Because $1 \in A$, we have $x \geq 1$. We use trichotomy to prove that $x^2 = 2$.

First suppose that $x^2 < 2$. The number $(2x + 1)/(2 - x^2)$ is positive. By the Archimedean Property, choose $n \in \mathbb{N}$ such that

$$n > \max \left\{ 1, \frac{2x + 1}{2 - x^2} \right\}.$$

Then $1/n < 1$ and $1/n < (2 - x^2)/(2x + 1)$. Consequently

$$\begin{aligned} \left(x + \frac{1}{n} \right)^2 &= x^2 + \frac{2x}{n} + \frac{1}{n^2} \\ &= x^2 + \frac{1}{n} \left(2x + \frac{1}{n} \right) \\ &< x^2 + \frac{1}{n} (2x + 1) \\ &< x^2 + (2 - x^2) \\ &= 2 \end{aligned}$$

Thus $x + 1/n \in A$, contradicting that x is an upper bound. Hence $x^2 < 2$ is impossible.

Next suppose that $x^2 > 2$. Choose $m \in \mathbb{N}$ with

$$m > \max \left\{ 1, \frac{2x}{x^2 - 2} \right\}.$$

Because $x \geq 1$ and $1/m < 1$, the number $y = x - 1/m$ is nonnegative. Also

$$\begin{aligned} y^2 &= \left(x - \frac{1}{m}\right)^2 = x^2 - \frac{2x}{m} + \frac{1}{m^2} \\ &> x^2 - \frac{2x}{m} \\ &> x^2 - (2x) \frac{x^2 - 2}{2x} \\ &= 2 \end{aligned}$$

We claim that y is an upper bound for A . If some $a \in A$ satisfied $a > y$, then $a, y \geq 0$ and Theorem 6.5 would give $a^2 > y^2 > 2$, contrary to $a \in A$. Therefore every element of A is at most y . But $y < x$, contradicting that x is the least upper bound. Hence $x^2 > 2$ is impossible.

Trichotomy leaves only $x^2 = 2$.

CHAPTER 7

INTRODUCTION TO CARDINALITY

INTRODUCTION

What do we mean when we say that there are *four* suits in a standard deck of cards? More generally, what does it mean to count any collection of objects? Since you may no longer actually think about counting, it may help to watch a child who is just learning to count. Given four objects to count, the child is likely to point to each object in turn and count “one, two, three, four.” In other words, the child is explicitly constructing a bijective function between the objects she is counting and the elements of the set $\{1, 2, 3, 4\}$. Our goal in this chapter is to extend this idea to infinite sets.

7.1 THE CARDINALITY OF A SET

For sets A and B , write $A \equiv B$ if there is a bijective function $f : A \rightarrow B$. In this case we say that A and B are *equinumerous* or that they have the same *cardinality*. Write $A \leq B$ if there is an injective function $f : A \rightarrow B$; equivalently, $B \geq A$. We write $A < B$ to indicate that $A \leq B$ and $A \not\equiv B$. It is also common to write $|A| = |B|$ rather than $A \equiv B$.

THEOREM 7.1.

For any set $\mathcal{C}$ whose elements are sets, equinumerosity restricts to an equivalence relation on $\mathcal{C}$.

Proof.

Let A , B , and C be arbitrary elements of $\mathcal{C}$. Since the identity function on any set is a bijection, $A \equiv A$ and $\equiv$ is reflexive.

If $A \equiv B$, then there is a bijection $f : A \rightarrow B$. By Theorem 5.26, its inverse $f^{-1} : B \rightarrow A$ is a function; it is bijective because an inverse interchanges the roles of inputs and outputs. Hence $B \equiv A$ and $\equiv$ is symmetric.

To see that $\equiv$ is transitive, suppose that $A \equiv B$ and $B \equiv C$. By definition there are bijective functions $f : A \rightarrow B$ and $g : B \rightarrow C$. Now apply Corollary 5.15 to see that $g \circ f : A \rightarrow C$ is a bijective function. Hence $A \equiv C$ and $\equiv$ is transitive. Therefore, $\equiv$ is an equivalence relation as desired.

Note that Theorem 7.1 allows us to say that two sets A and B have the same cardinality if we are able to find a bijection from A to B or a bijection from B to A .

THEOREM 7.2.

On any set $\mathcal{C}$ whose elements are sets, the relation $\leq$ is reflexive and transitive.

Proof.

Let A , B , and C be elements of $\mathcal{C}$. Since the identity function on any set is injective, $A \leq A$ and $\leq$ is reflexive. To see that $\leq$ is transitive, suppose that $A \leq B$ and $B \leq C$. By definition there are injections $f : A \rightarrow B$ and $g : B \rightarrow C$. We apply Theorem 5.14 to see that the composition $g \circ f : A \rightarrow C$ is also injective, hence $A \leq C$.

While we would not expect $\leq$ to be symmetric, we can ask for a cardinal analogue of antisymmetry. Mutual injections do not force two sets to be literally equal, but do they force the sets to be equinumerous? Georg Cantor (1845–1918) was interested in this question, and Felix Bernstein (1878–1956) proved that the answer is yes. The result is usually known as the Cantor–Bernstein Theorem.

THEOREM 7.3 (Cantor–Bernstein).

If $A \leq B$ and $B \leq A$, then $A \equiv B$.

We prove this theorem in Chapter A. The following example uses the same kind of chain-shifting idea to construct a bijection between the open interval $(-1, 1)$ and the closed interval $[-1, 1]$.

EXAMPLE 7.4.

Define the functions $f : (-1, 1) \rightarrow [-1, 1]$ and $g : [-1, 1] \rightarrow (-1, 1)$ by $f(x) = x$ and $g(x) = x/2$, respectively. It is easy to show that both of these functions are injective, so $(-1, 1) \leq [-1, 1]$ and $[-1, 1] \leq (-1, 1)$. The Cantor–Bernstein Theorem implies that there must be a bijection between the open interval $(-1, 1)$ and the closed interval $[-1, 1]$, but the theorem itself doesn't tell us how to find such a bijection. We construct such a bijection here.

We wish to find a bijective function $h : (-1, 1) \rightarrow [-1, 1]$. For most elements $x \in (-1, 1)$ we want $h(x) = f(x) = x$. Unfortunately, if we let $h(x) = f(x)$ for all x , then the function is not bijective because $-1 \neq f(x)$ and $1 \neq f(x)$ for all $x \in (-1, 1)$. We use the function g to help us fix this problem. Consider first the element $1 \in [-1, 1]$. While $1 \neq f(x)$ for any x , there is an element of the open interval associated with 1 by g . In particular $g(1) = 1/2$. We could define h so that $h(1/2) = 1$ and $h(x) = f(x)$ for other $x \in (-1, 1)$, but that creates a new problem. Now 1 is in the image of the function, but $1/2$ is not. To fix this we let $h(1/4) = 1/2$, because $g(1/2) = 1/4$. Of course, now $1/4$ is not in the image of h . We

continue fixing one problem at a time using g , each time creating a new problem. Naturally, we will need to do the same with -1 .

These particular functions are simple enough that we can write down a formula for the function h that we end up with. We end up defining $h : (-1, 1) \rightarrow [-1, 1]$ by

$$h(x) = \begin{cases} 2^{-(n-1)} & \text{if } x = 2^{-n} \text{ for some } n \in \mathbb{N} \\ -2^{-(n-1)} & \text{if } x = -2^{-n} \text{ for some } n \in \mathbb{N} \\ x & \text{otherwise} \end{cases}$$

We claim that this function is the desired bijection.

We first show that h is surjective. To see this, let $t \in [-1, 1]$. If $t = 2^{-(n-1)}$ for some $n \in \mathbb{N}$, then $h(2^{-n}) = t$. If $t = -2^{-(n-1)}$ for some $n \in \mathbb{N}$, then $h(-2^{-n}) = t$. For any other $t \in [-1, 1]$ we have $h(t) = t$. In any case $t = h(x)$ for some $x \in (-1, 1)$, so h is surjective.

To see that h is injective, suppose that $x \neq y$ are both elements of $(-1, 1)$. We consider several cases.

Case 1. $x = 2^{-n}$, $y = 2^{-k}$ for some $n, k \in \mathbb{N}$. Since $x \neq y$, it follows that $n \neq k$. Hence $n - 1 \neq k - 1$ and we have $h(x) = 2^{-(n-1)} \neq 2^{-(k-1)} = h(y)$.

Case 2. $x = -2^{-n}$, $y = -2^{-k}$ for some $n, k \in \mathbb{N}$. Since $x \neq y$, it follows that $n \neq k$. Hence $n - 1 \neq k - 1$ and we have $h(x) = -2^{-(n-1)} \neq -2^{-(k-1)} = h(y)$.

Case 3. $x = 2^{-n}$, $y = -2^{-k}$ for some $n, k \in \mathbb{N}$. In this case we have $h(x) = 2^{-(n-1)} \neq -2^{-(k-1)} = h(y)$.

Case 4. $x = -2^{-n}$, $y = 2^{-k}$ for some $n, k \in \mathbb{N}$. In this case we have $h(x) = -2^{-(n-1)} \neq 2^{-(k-1)} = h(y)$.

Case 5. $x = \pm 2^{-n}$ for some $n \in \mathbb{N}$ and $y \neq \pm 2^{-k}$ for any $k \in \mathbb{N}$. In this case we have $h(x) = \pm 2^{-(n-1)} \neq y = h(y)$.

Case 6. $y = \pm 2^{-n}$ for some $n \in \mathbb{N}$ and $x \neq \pm 2^{-k}$ for any $k \in \mathbb{N}$. In this case we have $h(x) = x \neq \pm 2^{-(n-1)} = h(y)$.

Case 7. $x \neq \pm 2^{-n}$ and $y \neq \pm 2^{-n}$ for any $n \in \mathbb{N}$. In this case we have $h(x) = x \neq y = h(y)$.

In all cases we have $h(x) \neq h(y)$, so h is injective.

7.2 FINITE SETS

QUESTION.

What does it mean to say that a set A is finite?

At first glance, this may seem like something you've known for a long time. Don't we just mean that we can count the elements of A ? If so, is the number of cells in your body finite? Can you count them? Maybe the answer to our question isn't quite so obvious after all.

For each $n \in \mathbb{N} \cup \{0\}$, define

$$I_0 = \emptyset \quad \text{and} \quad I_n = \{1, 2, \dots, n\} \quad \text{when } n \in \mathbb{N}.$$

Thus $I_4 = \{1, 2, 3, 4\}$. These sets let us formalize the idea of counting while handling the empty set without a separate convention.

DEFINITION 7.5.

Let A be a set and let $n \in \mathbb{N} \cup \{0\}$. We say that A has n elements if $A \equiv I_n$. The set A is *finite* if $A \equiv I_n$ for some $n \in \mathbb{N} \cup \{0\}$, and it is *infinite* if it is not finite.

THEOREM 7.6. *The set $\mathbb{N}$ is infinite.*

Proof.

If $\mathbb{N}$ were finite, it could be listed as $\{a_1, a_2, \dots, a_n\}$ for some $n \in \mathbb{N}$; the empty case is impossible. Every nonempty finite list of natural numbers has a largest entry, as follows by induction on its length. If M were the largest member of this list, then $M + 1 \in \mathbb{N}$ would be absent from the list, a contradiction.

Composing a bijection $A \rightarrow I_n$ with the inclusion $I_n \hookrightarrow \mathbb{N}$ gives the following.

THEOREM 7.7.

If A is finite, then $A \leq \mathbb{N}$.

Now that we have a definition of finite, let's reconsider the set C of cells in your body. Is C finite? If so, for which n is $C \equiv I_n$? We still can't really count them, so perhaps we've just obscured the question rather than answering it. Scientists estimate that there are about 10,000,000,000,000 cells in the average adult human body. That's not an actual count of

the number of cells in any individual human body, though. These kinds of estimates are based on the sizes of various kinds of cells and the approximate proportion of each kind of cell in the body. In fact, we could determine the maximum number of cells that might be in a person's body by figuring out how many of the smallest kinds of cells would be required to build a body of a particular volume, or weight, etc. It seems reasonable to think that we could say a set was finite if we were sure it had at most n elements for some natural number n . That is the intent of the next result.

THEOREM 7.8.

If $A \leq I_n$ for some $n \in \mathbb{N} \cup \{0\}$, then A is finite.

Suppose there is an injection $f : A \rightarrow I_n$. Our definition of finite asks for a bijection with some I_k , while f need not be surjective. The next lemma removes one unused point from the codomain at a time.

LEMMA 7.9.

Let $n \in \mathbb{N}$. If $f : A \rightarrow I_n$ is injective but not surjective, then there is an injection $g : A \rightarrow I_{n-1}$.

Proof.

Since f is not surjective, $B = I_n \setminus f[A]$ is nonempty. Choose $b \in B$. If $b = n$, then the range of f is already in I_{n-1} , so set $g = f$. If $b \neq n$, define

$$g(a) = \begin{cases} f(a), & f(a) \neq n, \\ b, & f(a) = n. \end{cases}$$

In either case g maps A into I_{n-1} . It remains to be shown that g is injective. Suppose that $a_1, a_2 \in A$ and that $g(a_1) = g(a_2) = m$. If $m = b$, then $b \notin f[A]$, so the first branch in the definition of g cannot produce b ; consequently $f(a_1) = n = f(a_2)$. If $m \neq b$, then the definition of g implies that $f(a_1) = m = f(a_2)$. In either case $f(a_1) = f(a_2)$, so $a_1 = a_2$ because f is injective. Therefore g is injective.

We will now prove the theorem.

Proof of Theorem 7.8.

If $A = \emptyset$, then $A \equiv I_0$ and the result is immediate. Assume $A \neq \emptyset$. The hypothesis then forces $n \geq 1$. Let $f_0 : A \rightarrow I_n$ be injective. If f_0 is surjective, it is the required bijection. Otherwise, Lemma 7.9 gives an injection $f_1 : A \rightarrow I_{n-1}$.

If f_1 is surjective, it is a bijection. If not, apply the lemma again to obtain $f_2 : A \rightarrow I_{n-2}$.

Continue. If one of the injections $f_i : A \rightarrow I_{n-i}$ is surjective, it is a bijection. The process must stop by $i = n - 1$: an injection from the nonempty set A into $I_1 = \{1\}$ is automatically surjective.

Thus $A \equiv I_{n-i}$ for some $0 \leq i \leq n - 1$, so A is finite.

Consequently, if $\mathbb{N} \leq A$, then A is infinite.

If A were finite, a bijection from A to some I_n , composed with an injection from $\mathbb{N}$ to A , would give an injection $\mathbb{N} \rightarrow I_n$. By Theorem 7.8, this would make $\mathbb{N}$ finite, contradicting Theorem 7.6.

We conclude this section with a question, the answer to which may seem obvious to you.

QUESTION 7.10.

If $m, n \in \mathbb{N} \cup \{0\}$ and $m \neq n$, can you prove that $I_m \not\cong I_n$?

We prove by induction on m the stronger fact that I_k cannot inject into I_m whenever $k > m$. For $m = 0$, the codomain is empty while the domain is nonempty, so the claim is immediate.

Assume the claim holds for m , let $k > m + 1$, and suppose that $f : I_k \rightarrow I_{m+1}$ is injective. If $m + 1$ is not in the range of f , then f is an injection into I_m , contrary to the inductive hypothesis. If $m + 1$ is in the range, remove its unique preimage from I_k and restrict f . After relabeling the remaining $k - 1$ elements as I_{k-1} , this gives an injection $I_{k-1} \rightarrow I_m$. Since $k - 1 > m$, this again contradicts the inductive hypothesis.

Thus no injection $I_n \rightarrow I_m$ exists when $m < n$. A bijection $I_m \rightarrow I_n$ would have an inverse of exactly that kind. If $n < m$, interchange the names of m and n .

7.3 DENUMERABLE SETS

Georg Cantor was able to define a complete system of infinite numbers and of arithmetic on those numbers. We will not discuss his system here, but we will look at one particular kind of infinite number that is important in many areas of mathematics.

THEOREM 7.11.

If A is an infinite set and B is a finite subset of A , then the set $A \setminus B$ is infinite.

Proof.

Suppose to the contrary that $A \setminus B$ is finite. Choose bijections $f : B \rightarrow I_m$ and $g : A \setminus B \rightarrow I_n$ for suitable $m, n \in \mathbb{N} \cup \{0\}$. Define $h : A \rightarrow I_{m+n}$ by

$$h(a) = \begin{cases} f(a), & a \in B, \\ m + g(a), & a \in A \setminus B. \end{cases}$$

The two branches have disjoint ranges, and each is injective, so h is injective. Theorem 7.8 would then make A finite, contrary to the hypothesis. Thus $A \setminus B$ is infinite.

DEFINITION 7.12.

Let A be a set. We say that A is:

- *denumerable* if $A \equiv \mathbb{N}$.
- *countable* if A is either finite or denumerable.
- *uncountable* if A is infinite and not denumerable.

EXAMPLE 7.13.

The set $A = \{2k \mid k \in \mathbb{N}\}$ of even natural numbers is denumerable. To see this, define the function $f : \mathbb{N} \rightarrow A$ by $f(n) = 2n$. It is routine to check that f is a bijection, so $\mathbb{N} \equiv A$ as desired.

The preceding example points out an important difference between finite and infinite sets. The set A is a proper subset of $\mathbb{N}$, but has the same cardinality as $\mathbb{N}$. A set equinumerous with one of its proper subsets is called *Dedekind-infinite*. To see why a finite set cannot have this property, suppose $A \equiv I_n$ and $B \subsetneq A$. The image of B under a bijection $A \rightarrow I_n$ is a proper subset S of I_n ; in particular, $n \geq 1$. The inclusion $S \rightarrow I_n$ is not surjective, so Lemma 7.9 gives an injection $S \rightarrow I_{n-1}$. If $B \equiv A$, this would produce an injection $I_n \rightarrow I_{n-1}$, contradicting the induction in the completion of Question 7.10. Therefore every Dedekind-infinite set is infinite. The converse holds in standard ZFC set theory, but its proof uses a form of the Axiom of Choice; without Choice, the two notions need not coincide.

We would like to determine which of our results about finite sets are also true for denumerable sets. If A and B are denumerable, must $A \cup B$ also be denumerable? Are subsets of denumerable sets denumerable? Is there an analog of Theorem 7.8? Are there other ways to tell that a set is denumerable?

THEOREM 7.14.

If A is a denumerable set and $B \equiv A$, then B is denumerable.

Proof. By definition we have $A \equiv \mathbb{N}$. Since $\equiv$ is an equivalence relation, it follows that $B \equiv \mathbb{N}$ as desired.

THEOREM 7.15.

If $A \subseteq \mathbb{N}$, then A is countable.

Proof.

If A is finite, then A is countable by definition. Assume that A is infinite. We define a function $f : \mathbb{N} \rightarrow A$ as follows. Recall that every nonempty subset of $\mathbb{N}$ has a smallest element. Let $f(1)$ be the smallest element of $A_0 = A$. Since A is infinite, the set $A_1 = A \setminus \{f(1)\}$ is nonempty, so we may define $f(2)$ to be the smallest element of A_1 . Continuing recursively, suppose that we have defined $f(1), \dots, f(n)$ for some $n \in \mathbb{N}$. Let $A_n = A \setminus \{f(1), \dots, f(n)\}$. Since $\{f(1), \dots, f(n)\}$ is finite, A_n is infinite by Theorem 7.11. In particular, A_n is nonempty and we may define $f(n+1)$ to be the smallest element of this set. Before showing that f is bijective we note the following facts that follow immediately from our construction:

1. For every natural number n , $A \setminus A_n = \{f(1), \dots, f(n)\}$.
2. For every natural number n , $f(n) \geq n$.
3. For all natural numbers m, n , if $f(n) \in A_m$ then $m < n$.
4. For all natural numbers $m < n$, $f(n) \in A_m$.
5. For all natural numbers $m \leq n$, $f(m) \notin A_n$.

For any two natural numbers $m < n$ we have shown that $f(n) \in A_m$ and $f(m) \notin A_m$, so $f(n) \neq f(m)$ and f is injective. To see that f is surjective, let $k \in A$. We must show that $k = f(m)$ for some $m \in \mathbb{N}$. If $k = f(k)$, we are done. Otherwise we have $f(k) > k$. Now $f(k)$ is the smallest element of A_{k-1} and $k < f(k)$, so $k \notin A_{k-1}$. We also know that $k \in A$, so it

follows that $k \in A \setminus A_{k-1} = \{f(1), \dots, f(k-1)\}$. Therefore $k = f(n)$ for some $1 \leq n < k$ and f is surjective as desired.

COROLLARY 7.16. *A set A is countable if and only if $A \leq \mathbb{N}$.*

Proof.

Suppose first that A is countable. If A is finite, then $A \leq \mathbb{N}$ by Theorem 7.7. If A is denumerable, a bijection from A to $\mathbb{N}$ is in particular an injection.

Conversely, suppose that $f : A \rightarrow \mathbb{N}$ is injective. After restricting its codomain, $f : A \rightarrow f[A]$ is a bijection, and Theorem 7.15 shows that $f[A]$ is countable. If $f[A]$ is finite, then A is finite; if $f[A]$ is denumerable, Theorem 7.14 shows that A is denumerable. Thus A is countable.

COROLLARY 7.17. *A subset of a countable set is countable.*

Proof.

Let $C \subseteq A$, where A is countable. The inclusion $C \rightarrow A$ is injective, and Corollary 7.16 supplies an injection $A \rightarrow \mathbb{N}$. Their composition is an injection $C \rightarrow \mathbb{N}$, so Corollary 7.16 shows that C is countable.

Thus a set A is countable exactly when there is an injection from A to $\mathbb{N}$. If A is nonempty, this is also equivalent to finding a surjection from $\mathbb{N}$ onto A . In one direction, invert an injection on its range and send unused natural numbers to a fixed element of A ; in the other, assign each $a \in A$ the least natural number mapping to it. The nonempty hypothesis is necessary: there is no function from $\mathbb{N}$ onto $\emptyset$.

COROLLARY 7.18.

A set A is countable if either of the following is true:

- (i) *There is an injection $f : A \rightarrow \mathbb{N}$.*
- (ii) *$A \neq \emptyset$ and there is a surjection $f : \mathbb{N} \rightarrow A$.*

THEOREM 7.19.

The union of two denumerable sets is denumerable.

Proof.

Suppose that we are given any two denumerable sets A and B . By definition there are bijective functions $f : A \rightarrow \mathbb{N}$ and $g : B \rightarrow \mathbb{N}$. We define a function $h : A \cup B \rightarrow \mathbb{N}$ by the following rule:

$$h(x) = \begin{cases} 2f(x) & \text{if } x \in A \\ 2g(x) + 1 & \text{if } x \notin A \end{cases}$$

We claim that h is an injection. To see this, let $x \neq y$ be two elements of $A \cup B$. If $x \in A$ and $y \notin A$, then $h(x) \neq h(y)$ since $h(x)$ is even and $h(y)$ is odd. Similarly if $x \notin A$ and $y \in A$, then $h(x) \neq h(y)$. If x and y are both in A , then $h(x) = 2f(x)$ and $h(y) = 2f(y)$, so $h(x) \neq h(y)$ because f is injective. Finally, if neither x nor y are in A , then $h(x) = 2g(x) + 1$ and $h(y) = 2g(y) + 1$, so $h(x) \neq h(y)$ because g is injective. In any case, we have shown that $h(x) \neq h(y)$ so $h : A \cup B \rightarrow \mathbb{N}$ is injective.

We have shown that $A \cup B \leq \mathbb{N}$, so $A \cup B$ is countable. It is actually denumerable: the inclusion $A \hookrightarrow A \cup B$ is injective, and $\mathbb{N} \leq A$ because A is denumerable. Transitivity gives $\mathbb{N} \leq A \cup B$. Cantor–Bernstein now yields $A \cup B \equiv \mathbb{N}$.

Using Mathematical Induction on the number of sets, we obtain:

COROLLARY 7.20.

The union of any positive finite number of denumerable sets is denumerable.

COROLLARY 7.21. *The union of two countable sets is countable.*

Proof.

Let A and B be countable. By Corollary 7.16, choose injections $f : A \rightarrow \mathbb{N}$ and $g : B \rightarrow \mathbb{N}$. Define

$$h(x) = \begin{cases} 2f(x), & x \in A, \\ 2g(x) + 1, & x \in B \setminus A. \end{cases}$$

The two branches have disjoint ranges and each branch is injective, so $h : A \cup B \rightarrow \mathbb{N}$ is injective. Corollary 7.16 now shows that $A \cup B$ is countable.

THEOREM 7.22.

The set $\mathbb{N} \times \mathbb{N}$ is denumerable.

Proof.

The map $n \mapsto (n, 1)$ injects $\mathbb{N}$ into $\mathbb{N} \times \mathbb{N}$, so the consequence following Theorem 7.8 shows that the product is infinite. It therefore suffices to prove that it is countable. Define the function $f : \mathbb{N} \times \mathbb{N} \rightarrow \mathbb{N}$ by $f(a, b) = 2^{a-1}(2b - 1)$. We will show that f is injective, then apply Corollary 7.18. To do so, suppose that

$$2^{a-1}(2b - 1) = 2^{c-1}(2d - 1) \quad (7.1)$$

for $(a, b), (c, d) \in \mathbb{N} \times \mathbb{N}$. We show first that $a = c$. If not, then we may assume without loss of generality that $a < c$. Dividing both sides of Equation (7.1) by 2^{a-1} yields $(2b - 1) = 2^{c-a}(2d - 1)$. But this is a contradiction since the quantity on the left side of the equation is odd and the quantity on the right is even. Since $a = c$, we may reduce Equation (7.1) to $(2b - 1) = (2d - 1)$, from which it follows that $b = d$. Now $(a, b) = (c, d)$ and f is injective as desired.¹

¹While it is not necessary for the purposes of this example, it is not too difficult to show that the function f defined here is actually bijective.

7.3.1 THE RATIONAL NUMBERS

At first glance it may seem that there are more rational numbers than there are natural numbers. After all, there are infinitely many rational numbers between any two natural numbers. One of Cantor's accomplishments was to show that the set of rational numbers is actually denumerable. Our intuition developed from years of working with finite sets just doesn't serve us very well when working with infinite sets.

LEMMA 7.23.

The set $\mathbb{Q}^+$ of positive rational numbers is countable.

Proof.

Let $F = \mathbb{N} \times \mathbb{N}$. We may write a pair $(a, b) \in F$ as the formal fraction a/b , but distinct pairs remain distinct even when they represent the same rational number. By Theorem 7.22, F is denumerable. Define

$$p : F \rightarrow \mathbb{Q}^+, \quad p(a, b) = \frac{a}{b},$$

where the expression on the right is now interpreted as a rational number. Every positive rational number has such a representation, so p is surjective. Choose a bijection $e : \mathbb{N} \rightarrow F$. Then $p \circ e : \mathbb{N} \rightarrow \mathbb{Q}^+$ is surjective, and Corollary 7.18 shows that $\mathbb{Q}^+$ is countable.

Since the function taking every positive rational to its additive inverse is bijective, the following corollary is immediate from the lemma.

COROLLARY 7.24.

The set $\mathbb{Q}^-$ of negative rational numbers is countable.

We know that $\mathbb{Q}^+$, $\mathbb{Q}^-$, and $\{0\}$ are all countable sets. Applying Corollary 7.21 twice, we conclude:

THEOREM 7.25.

The set $\mathbb{Q}$ of rational numbers is denumerable.

Proof.

The preceding results show that $\mathbb{Q}$ is countable. The inclusion $\mathbb{N} \hookrightarrow \mathbb{Q}$, together with the consequence following Theorem 7.8, shows that $\mathbb{Q}$ is infinite. Therefore $\mathbb{Q}$ is denumerable.

7.3.2 THE REAL NUMBERS

By this point it may seem possible that all sets are countable, making denumerability a useless distinction. We will show that this is not true by proving that the real numbers are uncountable. A real number can have two decimal expansions; for example, $1.000 \dots = 0.999 \dots$. To make decimal digits unambiguous, we choose for each real number its *canonical* expansion: the one that is not eventually all 9s.

THEOREM 7.26.

The set $\mathbb{R}$ of real numbers is uncountable.

Proof.

Suppose to the contrary that $\mathbb{R}$ is countable. Since $\mathbb{R}$ is nonempty, there would be a surjection $f : \mathbb{N} \rightarrow \mathbb{R}$. Let $f(n)_n$ denote the n th digit after the decimal point in the canonical expansion of $f(n)$. Define digits

$$y_n = \begin{cases} 2, & \text{if } f(n)_n = 1, \\ 1, & \text{if } f(n)_n \neq 1, \end{cases}$$

and let $y = 0.y_1y_2y_3\dots$. Because every y_n is 1 or 2, this expansion is canonical and represents a number $y \in (0, 1)$. For every $n \in \mathbb{N}$, the n th digit of y differs from the n th digit of $f(n)$. Thus $y \neq f(n)$ for every n , contradicting surjectivity.

APPENDIX A

THE CANTOR–BERNSTEIN THEOREM

The relation $X \leq Y$ means that X injects into Y . It is reflexive and transitive, but it is not literally antisymmetric on sets: distinct sets can have the same cardinality. The Cantor–Bernstein Theorem supplies the appropriate cardinal version of antisymmetry:

$$X \leq Y \text{ and } Y \leq X \implies X \equiv Y.$$

The proof below is constructive once the two injections are given.

A.1 THE FORWARD-CHAIN CONSTRUCTION

Assume that $f : X \rightarrow Y$ and $g : Y \rightarrow X$ are injective. Some points of X have no predecessor under g ; begin a forward chain at each such point. Define

$$A_0 = X \setminus g[Y], \quad A_{n+1} = g[f[A_n]] \quad (n = 0, 1, 2, \dots),$$

and let

$$A = \bigcup_{n=0}^{\infty} A_n = \{x \in X \mid x \in A_n \text{ for some } n \in \mathbb{N} \cup \{0\}\}.$$

On the chains in A we use f in the forward direction. Off those chains, every point has a unique predecessor under g , so we use that predecessor.

Proof of Theorem 7.3.

For $x \in X \setminus A$, we have $x \notin A_0$, hence $x \in g[Y]$. Thus there is a $y \in Y$ with $g(y) = x$, and this y is unique because g is injective. We may therefore define $h : X \rightarrow Y$ by

$$h(x) = \begin{cases} f(x), & x \in A, \\ \text{the unique } y \in Y \text{ such that } g(y) = x, & x \notin A. \end{cases}$$

We first prove that h is injective. Suppose $h(x_1) = h(x_2)$.

If $x_1, x_2 \in A$, then $f(x_1) = f(x_2)$, so $x_1 = x_2$ because f is injective.

If $x_1, x_2 \notin A$, then

$$x_1 = g(h(x_1)) = g(h(x_2)) = x_2.$$

It remains to rule out a mixed case. Suppose $x_1 \in A$ and $x_2 \notin A$. Choose $n \geq 0$ such that $x_1 \in A_n$. If $h(x_1) = h(x_2)$, then

$$x_2 = g(h(x_2)) = g(h(x_1)) = g(f(x_1)).$$

Hence $x_2 \in g[f[A_n]] = A_{n+1} \subseteq A$, a contradiction. The other mixed case is symmetric. Therefore h is injective.

To prove surjectivity, let $y \in Y$. If $g(y) \notin A$, then the second clause in the definition gives

$$h(g(y)) = y.$$

Now suppose $g(y) \in A$. Since $g(y) \in g[Y]$, it cannot lie in $A_0 = X \setminus g[Y]$. Therefore $g(y) \in A_{n+1}$ for some $n \geq 0$. By the definition of A_{n+1} , there is an $x \in A_n$ such that

$$g(y) = g(f(x)).$$

Injectivity of g gives $y = f(x)$, and because $x \in A$, $h(x) = f(x) = y$. Thus every $y \in Y$ has a preimage under h , so h is surjective. The function h is both injective and surjective; hence it is the desired bijection $X \rightarrow Y$.

A.2 WHY THE CONSTRUCTION WORKS

The set A_0 records where the backward search through g must stop. Each A_{n+1} advances those starting points one step through f and then g . Using f on every such forward chain prevents a missing predecessor from causing a gap. Everywhere else, the g -chains extend backward indefinitely, so reversing g is safe. The argument also covers empty sets without a separate case.